The goal is to make these steps execute quickly. Since steps 6-12 will be repeated many times (once for each coefficient in the filter), special attention must be given to these operations. Traditional microprocessors must generally carry out these 14 steps in *serial* (one after another), while DSPs are designed to perform them in *parallel*. In some cases, all of the operations within the loop (steps 6-12) can be completed in a *single clock cycle*. Let's look at the internal architecture that allows this magnificent performance.

Architecture of the Digital Signal Processor

One of the biggest bottlenecks in executing DSP algorithms is transferring information to and from memory. This includes *data*, such as samples from the input signal and the filter coefficients, as well as *program instructions*, the binary codes that go into the program sequencer. For example, suppose we need to multiply two numbers that reside somewhere in memory. To do this, we must fetch three binary values from memory, the numbers to be multiplied, plus the program instruction describing what to do.

Figure 28-4a shows how this seemingly simple task is done in a traditional microprocessor. This is often called a **Von Neumann architecture**, after the brilliant American mathematician John Von Neumann (1903-1957). Von Neumann guided the mathematics of many important discoveries of the early twentieth century. His many achievements include: developing the concept of a stored program computer, formalizing the mathematics of quantum mechanics, and work on the atomic bomb. If it was new and exciting, Von Neumann was there!

As shown in (a), a Von Neumann architecture contains a single memory and a single bus for transferring data into and out of the central processing unit (CPU). Multiplying two numbers requires at least three clock cycles, one to transfer each of the three numbers over the bus from the memory to the CPU. We don't count the time to transfer the result back to memory, because we assume that it remains in the CPU for additional manipulation (such as the sum of products in an FIR filter). The Von Neumann design is quite satisfactory when you are content to execute all of the required tasks in serial. In fact, most computers today are of the Von Neumann design. We only need other architectures when very fast processing is required, and we are willing to pay the price of increased complexity.

This leads us to the **Harvard architecture**, shown in (b). This is named for the work done at Harvard University in the 1940s under the leadership of Howard Aiken (1900-1973). As shown in this illustration, Aiken insisted on separate memories for data and program instructions, with separate buses for each. Since the buses operate independently, program instructions and data can be fetched at the same time, improving the speed over the single bus design. Most present day DSPs use this dual bus architecture.

Figure (c) illustrates the next level of sophistication, the **Super Harvard Architecture**. This term was coined by Analog Devices to describe the